**PRAKTIKUM BASIS DATA**

**MODUL 6**

**Scripting Subquery & Nested Subquery**

Disusun oleh:

Aradea Satria Permana (103122400014)

**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

**FAKULTAS INFORMATIKA**

**UNIVERSITAS TELKOM**

**APRIL 2026**

# JURNAL MODUL 6

1. Tampilkan nama dan umur member yang memesan teater dimana harganya lebih tinggi dari ‘Teater 1’ dan berjudul ‘Habibie AInun 3’! (Gunakan Subquery)

```
SELECT m.nama,
       FLOOR(MONTHS_BETWEEN(SYSDATE, m.tgl_lahir)/12) AS umur
FROM MEMBER m
JOIN TRANSAKSI t ON m.id_member = t.id_member
JOIN JADWAL j ON t.id_jadwal = j.id_jadwal
JOIN TEATER tr ON j.id_theater = tr.nomor_teater
JOIN FILM f ON j.id_film = f.id_film
WHERE f.judul = 'Habibie Ainun 3';
```

All Rows Fetched: 6 in 0.031 seconds

| | NAMA | UMUR |
|---|---|---|
| 1 | Budi | 41 |
| 2 | Ari | 42 |
| 3 | Fahmi | 39 |
| 4 | Doni | 39 |
| 5 | Dono | 35 |
| 6 | Joko | 37 |

2. Tampilkan film yang ditayangkan pada teater paling murah atau paling mahal, urutkan sesuai harga! (Gunakan Subquery)

```
SELECT f.judul, tr.nomor_teater, tr.harga
FROM TRANSAKSI t
JOIN JADWAL j ON t.id_jadwal = j.id_jadwal
JOIN TEATER tr ON j.id_theater = tr.nomor_teater
JOIN FILM f ON j.id_film = f.id_film
WHERE tr.harga = (SELECT MIN(harga) FROM TEATER)
   OR tr.harga = (SELECT MAX(harga) FROM TEATER)
ORDER BY tr.harga;
```

All Rows Fetched: 5 in 0.003 seconds

| | JUDUL | NOMOR_TEATER | HARGA |
|---|---|---|---|
| 1 | Keluarga Cemara | Teater 1 | 30000 |
| 2 | Keluarga Cemara | Teater 1 | 30000 |
| 3 | Keluarga Cemara | Teater 1 | 30000 |
| 4 | Keluarga Cemara | Teater 1 | 30000 |
| 5 | Keluarga Cemara | Teater 5 | 70000 |

3. Lakukan normalisasi pada skema relasi berikut sehingga memenuhi syarat sampai Normal Bentuk Ketiga!

```
SELECT f.judul, COUNT(*) AS jumlah_tayang
FROM TRANSAKSI t
JOIN JADWAL j ON t.id_jadwal = j.id_jadwal
JOIN FILM f ON j.id_film = f.id_film
GROUP BY f.judul
HAVING COUNT(*) > (
    SELECT MIN(jml)
    FROM (
        SELECT COUNT(*) AS jml
        FROM TRANSAKSI t2
        JOIN JADWAL j2 ON t2.id_jadwal = j2.id_jadwal
        GROUP BY j2.id_film
    )
);
```

All Rows Fetched: 1 in 0.015 seconds

| | JUDUL | JUMLAH_TAYANG |
|---|---|---|
| 1 | Habibie Ainun 3 | 6 |

4. Tampilkan nomor teater dan film yang ditayangkan melebihi rata-rata penayangan juga berdurasi lebih dari 100! (Gunakan Subquery)

```
FROM TRANSAKSI t
JOIN JADWAL j ON t.id_jadwal = j.id_jadwal
JOIN TEATER tr ON j.id_theater = tr.nomor_teater
JOIN FILM f ON j.id_film = f.id_film
GROUP BY tr.nomor_teater, f.judul, f.durasi
HAVING COUNT(*) > (
    SELECT AVG(jml)
    FROM (
        SELECT COUNT(*) AS jml
        FROM TRANSAKSI t2
        JOIN JADWAL j2 ON t2.id_jadwal = j2.id_jadwal
        GROUP BY j2.id_theater
    )
)
AND f.durasi > 100;
```

All Rows Fetched: 1 in 0.032 seconds

| | NOMOR_TEATER | JUDUL | JUMLAH_TAYANG |
|---|---|---|---|
| 1 | Teater 1 | Keluarga Cemara | 4 |

5. Tampilkan nama member dan nomor hp beserta film yang ditonton, dimana dia menonton pada teater yang berharga lebih dari ‘30000’ dan member memiliki no hp berawalan ‘081’! (Gunakan Subquery)

modul8.sql | Modul6

Worksheet | Query Builder

```sql
SELECT m.nama, m.no_hp, f.judul
FROM MEMBER m
JOIN TRANSAKSI t ON m.id_member = t.id_member
JOIN JADWAL j ON t.id_jadwal = j.id_jadwal
JOIN TEATER tr ON j.id_theater = tr.nomor_teater
JOIN FILM f ON j.id_film = f.id_film
WHERE tr.harga > 30000
AND m.no_hp LIKE '081%';
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.017 seconds

| | NAMA | NO_HP | JUDUL |
|---|---|---|---|
| 1 | Budi | 081367589632 | Habibie Ainun 3 |
| 2 | Ari | 081267867543 | Habibie Ainun 3 |
| 3 | Doni | 081398426789 | Habibie Ainun 3 |
| 4 | Dono | 081287234567 | Habibie Ainun 3 |
| 5 | Joko | 081223670942 | Habibie Ainun 3 |